

Introduction

RSA (Rivest–Shamir–Adleman) is a widely-used public-key cryptosystem based on the difficulty of factoring large integers. Since its introduction in 1977, RSA has been foundational in digital security, enabling encrypted communication, digital signatures, and secure data exchange. This report investigates RSA's real-world applications, quantum-era threats, and practical implementation.

❖ Phase 1: RSA in the Real World

1. Secure Websites (HTTPS/TLS)

RSA is used in Transport Layer Security (TLS) to protect HTTPS connections. During a TLS handshake:

- The client requests a secure connection.
- The server sends its digital certificate, which includes its RSA public key.
- The client encrypts a symmetric session key with the server's public key.
- The server decrypts it using its private key.
- Both now use the symmetric session key for secure communication.

This ensures confidentiality, integrity, and authentication.

2. Email Encryption (PGP)

Pretty Good Privacy (PGP) uses RSA in combination with symmetric encryption:

- The message is encrypted using a symmetric key (e.g., AES).
- The symmetric key is encrypted with the recipient's RSA public key.
- The recipient decrypts the symmetric key using their private key to access the message.

RSA guarantees that only the intended recipient can decrypt the content.

3. Digital Signatures for Software Updates

RSA enables secure code signing:

- A hash of the software update is signed with the developer's private RSA key.
- The user verifies the signature using the developer's public key.

This process ensures the update's origin and integrity.

RSA in TLS and Blockchain

In addition to TLS, RSA is sometimes used in blockchain for identity verification and off-chain communication. While newer systems prefer elliptic curve cryptography, RSA remains relevant in legacy and enterprise systems.

Illustrations: Diagrams depicting RSA in TLS and digital signatures in blockchain were used to clarify integration.

❖ **Phase 2: Post-Quantum Threats to RSA**

1. **Shor's Algorithm and Quantum Threats**

Shor's algorithm can factor large integers in polynomial time using a quantum computer. Since RSA relies on the difficulty of factoring, Shor's algorithm poses a fundamental threat:

- It can derive the private key from the public key.
- It breaks RSA-based encryption and signatures.

RSA is therefore insecure in a post-quantum world.

2. **Post-Quantum Alternatives**

New algorithms are being developed to resist quantum attacks. Two examples include:

- **CRYSTALS-Kyber:** Lattice-based encryption using the MLWE problem; fast, quantum-safe, and selected by NIST.
- **NTRU:** Uses polynomial rings and hard lattice problems; efficient and well-established.

Comparison Table:

Feature	RSA	CRYSTALS-Kyber	NTRU
Security Basis	Integer factorization	Lattice (MLWE)	Lattice
Quantum Resistance	✗ No	✓ Yes	✓ Yes
Key Size	Large	Moderate	Moderate
Speed	Slower	Fast	Fast

Conclusion (Quantum Phase)

RSA’s vulnerability to quantum computing requires migration to post-quantum cryptography. Transitioning to algorithms like CRYSTALS-Kyber and NTRU is vital for future-proof security.

❖ Phase 3: RSA Test Case

Given:

- $P = 13, q = 17, e = 5, M = 19$

Step 1: Compute RSA Parameters

- $n = pq = 13 \times 17 = 221$
- $\phi(n) = (p - 1)(q - 1) = 12 \times 16 = 192$

Find d such that $ed \equiv 1 \pmod{192}$

- $5d \equiv 1 \pmod{192} \rightarrow d = 77$

Step 3: Encrypt Message

- $C = M^e \pmod n = 19^5 \pmod{221} = 143$

Step 4: Decrypt Ciphertext

$$M = C^d \bmod n = 143^{77} \bmod 221 = 19$$

Result:

- Public Key: (5, 221)
- Private Key: (77, 221)
- Ciphertext: 143
- Decrypted Message: 19

❖ Conclusion

RSA remains essential in current cybersecurity applications. It supports encrypted communication, authentication, and data integrity. However, quantum computing poses a serious threat to its long-term viability. The cryptography community must embrace post-quantum standards to ensure security in the future.

❖ References

- RSA Laboratories – PKCS #1
- Zimmermann, Phil. “PGP User’s Guide.” MIT Press.
- RFC 8446 – TLS Protocol
- Shor, P. “Algorithms for Quantum Computation.” (1994)
- NIST PQC Project
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